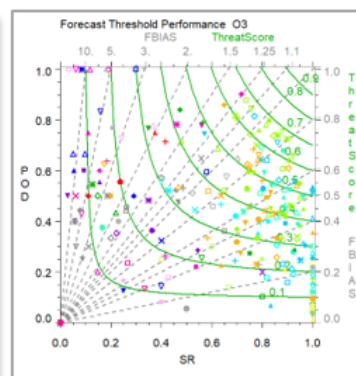
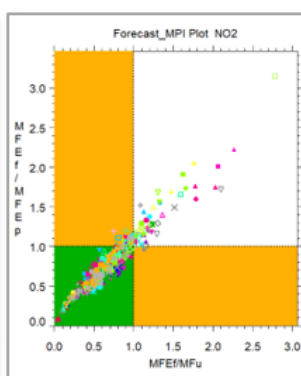
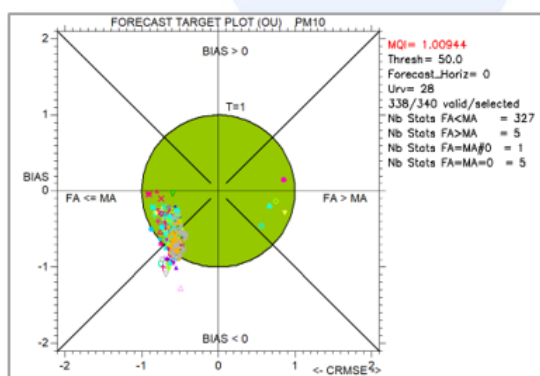


FAIRMODE Guidance

Document on Quality control indicators for AQ forecasts

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Contents

Abstract	2
Acknowledgements	3
Authors	3
Introduction	4
1. Forecasting indicators: rationale and methodology	5
1.1. The forecast Modelling Quality Indicator	5
1.2. The forecast Modelling Performance Indicators	6
1.3. Threshold exceedances indicators	7
1.4. Assessment of capability to predict air quality indices	9
2. Forecasting indicators: graphical implementation proposal	10
2.1. Forecast Target Plot	10
2.2. Forecast MPI Plot	11
2.3. Forecast Threshold Performance Plot	11
2.4. Diagrams for exceedance indicators	13
2.5. Forecast Summary Report	13
2.6. Forecast diagrams using Air Quality Indices	15
2.7. Overview of the metrics	16
3. Conclusions and open issues	18
4. References	19
List of abbreviations and definitions	20
List of figures	22
List of tables	23

Abstract

A standardized methodology for the validation of short-term air quality forecast applications was developed in the framework of the Forum for Air quality Modeling in Europe (FAIRMODE¹) activities. The proposed approach, focusing on specific indicators to be reviewed when evaluating a forecasting application, investigates the models' capability to detect sudden changes in pollutant concentration levels, predict threshold exceedances and reproduce air quality indices as well as overall model performance in relation to measured concentrations.

The proposed formulation relies on the definition of specific forecast modelling quality objectives and performance criteria, defining the minimum level of quality to be achieved by a forecasting application when it is used for policy purposes. The persistence model, which uses an appropriate recently observed value as the predicted value, is used as a benchmark for the definition of the Forecast Modelling Quality Indicator MQI_f (ratio between the root mean square error of the forecast model and the root mean square error of the persistence model). The modelling quality objective (MQO_f) requires MQI_f to be less than or equal to 1. To gain further insights on the strengths and weaknesses of the forecast application, additional modelling performance indicators (MPIs) are proposed as part of the evaluation protocol, based on the Mean Fractional Error (MFE).

Concerning the capability of the forecast modelling application in correctly providing categorical air quality forecasts, namely the exceedances of concentration thresholds, the most used threshold indicators can be applied. A fitness for purpose criterion was not definitively set within the proposed protocol.

The last model capability assessed within the proposed validation protocol concerns the correct prediction of air quality indices (AQI), designed to provide citizens with effective and simple information about air quality and its impact on their health. The current approach is based on a cumulative analysis of relative distributions of observed and forecasted AQIs. A fitness for purpose criterion was not definitively set within the proposed protocol.

Graphical implementations are proposed for the aforementioned indicators (Forecast Target Plot and MPI Plot, Forecast Threshold Performance Plot, Summary report, Forecast diagrams using Air Quality Indices). The aim of this document is to present the state of the art of the protocol so that it can be tested by the scientific community, especially with respect to targeting air quality forecasting services supporting policymaking in European Member States. Open issues are presented at the end of the document. The discussion on the proposed approach will continue within the FAIRMODE community, and upgrades and improvements of the current validation protocol will be fostered by its usage.

¹ The Forum for Air quality Modelling (FAIRMODE) is an initiative to bring together air quality modellers and users in order to promote and support the harmonised use of models by EU Member States, with emphasis on model application under the European Air Quality Directives.

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Introduction

Air pollution models play a key role in improving the scientific understanding of atmospheric processes and supporting policy decisions aimed at reducing human exposure to air pollution. The revised European Ambient Air Quality Directive 2024/2881/EC emphasises the use of models to support its implementation. According to 2024/2881/EC AAQD, forecasting tools are expected to support the prediction of exceedances and the provision of information to the public [Article 15 and Annex X (2-3)]. In cases of exceedance risk, forecasting tools are also expected to support Member States and local authorities in charge for air quality in establishing short-term action plans to reduce the risk or duration of such exceedances [Article 20].

A thorough assessment of model performance is fundamental to build confidence in model capabilities and is particularly critical when models are used to support policymaking. The objective of a model evaluation process is to prove that the performances are satisfactory for the model application's intended use, in other words, that it is "fit for purpose".

Over recent years, significant contributions to the definition of evaluation protocols and performance criteria have been made through the activities and coordination efforts of FAIRMODE. An evaluation protocol has been proposed/developed for the evaluation of model applications for regulatory air quality assessment (Thunis et al., 2012a, b; Pernigotti et al., 2013; Thunis et al., 2013; Janssen and Thunis, 2022).

More recently, the FAIRMODE discussion has expanded to include the development and testing of additional quality control indicators for evaluating forecast applications. An evaluation approach, focusing on the model ability to accurately predict sudden changes in the pollutant concentration levels and based on the usage of the persistence model as a benchmark (e.g. Mittermaier, 2008), has been proposed.

This standardized approach was first presented in the previous *FAIRMODE Guidance Document on Modelling Quality Objectives and Benchmarking* (Janssen and Thunis, 2022). After its publication (Piersanti et al., 2022; Vitali et al., 2023), the methodology is now described in detail in this stand-alone guidance document. Section 2 outlines the rationale and main features of the methodology. Section 3 presents examples of possible graphical implementations of the proposed procedures. Finally, Section 4 provides conclusions and discusses open issues and possible directions for further development.

1. Forecasting indicators: rationale and methodology

This section describes the main features of the proposed protocol. The main objective is to offer a common standardised template to facilitate the screening and comparison of forecast results.

The proposed approach focuses on the model's capability to:

1. detect sudden changes in concentration levels
2. predict threshold exceedances
3. reproduce air quality indices.

Note that the proposed approach is not exhaustive. It does not evaluate all relevant features of a forecast application, and other analyses will be helpful to gain further insights into the behaviour, strengths and shortcomings of an air quality forecast modelling application.

The methodology currently supports the following pollutants and metrics: the NO₂ daily maximum, O₃ daily maximum of 8 h moving average, and PM10 and PM2.5 daily means. Note that only one value per day is used and that no evaluation of the entire hourly time profile is made.

1.1. The forecast Modelling Quality Indicator

Predicting the status of air quality is useful in relation to reducing or preventing health impacts from acute air pollution episodes and to trigger short-term action plans. Therefore, assessing forecast application ability to accurately predict sudden changes in the pollutant concentration levels is of great importance. In the proposed protocol, this capability is assessed using the persistence model as a benchmark. The persistence model, by definition, cannot capture changes in concentration levels since it uses measurement data from the previous day as the forecast for the following days. The persistence approach, being the simplest method for prediction when no other information is available, is commonly used as a reference in weather forecast verification. Indeed, the persistence approach is the simplest method for predicting future behaviour if no other information is available and is often used as a point of reference in verifying the performances of weather forecasts (e.g. Knaff and Landsea, 1997; Mittermaier, 2008).

Within the proposed forecasting evaluation approach, the root mean square error (RMSE) of the forecast model is compared with the root mean square error of the persistence model. The forecast modelling quality indicator for forecast (MQI_f) is defined as the ratio between the two RMSEs; i.e.

$$MQI_f = \sqrt{\frac{\frac{1}{N} \sum_{i=1}^N (M_i - O_i)^2}{\frac{1}{N} \sum_{i=1}^N (P_i - O_i)^2}} \quad \text{and} \quad (1)$$

MQO_f is fulfilled if $MQI_f \leq 1$,

where M_i, P_i and O_i represent the forecast, persistence and measured values, respectively, for day i, and N is the number of days included in the time series. It is clear from the formula that MQI_f becomes 1 when the persistence model P is used as forecast model M. MQI_f values lower than 1 indicate better capabilities than the persistence model, whereas values larger than 1 indicate poorer performances.

The persistence model uses the observations from the previous day as an estimate for all forecast days. As an example, we can consider a 3-day forecast, providing concentration values for today (day0), tomorrow (day1) and the day after tomorrow (day2). For instance, if today is 5 February, the persistence model uses data referring to yesterday (4 February) for all forecast data produced

today. So, P_i refers to O_{i-1} for day0 (5 February), O_{i-2} for day1 (6 February) and O_{i-3} for day2 (7 February). More generally, the persistence model is related to the forecast horizon (FH=0, 1, 2, etc.) as follows:

$$P_i = O_{i-1-FH} \pm U(O_{i-1-FH}) \quad (2)$$

where the measurement uncertainty U is also considered, in accordance with the FAIRMODE approach. The methodology for estimating the measurement uncertainty as a function of the concentration values is described in Janssen and Thunis (2022).

MQI_f is computed for each monitoring point. The MQI_f associated with the 90th percentile worst station is used as the main indicator in the proposed benchmarking procedure. A forecasting application is considered fit for purpose if $MQI_f \leq 1$ for at least 90 % of available stations.

The required minimum number of stations and the recommended approaches to consider the 90th percentile in case of few stations are addressed in Janssen and Thunis (2022).

1.2. The forecast Modelling Performance Indicators

The MQI_f formulation combines two aspects: (1) checking how well the model prediction compares with measurements and (2) checking whether the model prediction performs better than a given benchmark (here the persistence model). Because MQI_f depends on the benchmark's skill, additional performance indicators (MPIs) are proposed to provide complementary insight.

These MPIs are based on the Mean Fractional Error (MFE), a normalized statistical indicator widely used in the literature, defined as follows:

$$MFE = \frac{2}{N} \sum_{i=1}^N \frac{|M_i - O_i|}{(M_i + O_i)} \quad (3)$$

MFE is chosen because it is a normalized error. It helps in interpreting the outcomes, since it does not depend on the magnitude of the corresponding concentration values. Moreover, MFE has been already used in the framework of FAIRMODE activities and performance criteria and goals, based on this indicator, have been defined in literature for PM (Boylan and Russell, 2006) and O_3 (Chemel et al., 2010).

Based on this indicator, two different MPIs are formulated to gain further insights.

1. Consistently with MQI_f formulation, an indicator is constructed to compare MFE_f from the forecast model to the MFE_p from the persistence model.
2. Forecast performances are also evaluated, regardless of persistence aspects, using an acceptability threshold based on measurement uncertainty. More in detail MFE_f is compared with the Mean Fractional Uncertainty (MF_U), defined as follows

$$MF_U = \frac{1}{N} \sum_{i=1}^N \frac{2U(O_i)}{O_i} \quad (4)$$

The definition of MF_U as the acceptability threshold derives from considering

$$MFE = \frac{2}{N} \sum_{i=1}^N \frac{|M_i - O_i|}{(M_i + O_i)} \sim \frac{1}{N} \sum_{i=1}^N \frac{|M_i - O_i|}{O_i} \quad (5)$$

and then setting the condition $|M_i - O_i| \leq 2U(O_i)$ consistently with the assumptions described in Janssen and Thunis (2022).

Features 1 and 2 turn into formulas (6) and (7) in Table 1, respectively. For both MPIs, modelling performance criteria (MPC) are proposed being fulfilled when MPIs are less than or equal to 1.

Table 1. Model performance indicators and criteria for MFE.

MPI	Equation	MPC
$MPI_1 = \frac{MFE_f}{MFE_p}$	(6)	MPC: MPIs ≤ 1
$MPI_2 = \frac{MFE_f}{MF_U}$	(7)	

Source: JRC

Using uncertainty parameters set in Janssen and Thunis (2022), double relative uncertainties $2U(O_i)/O_i$ as a function of concentration values show larger values in the low concentration range and then tend towards a constant value at higher concentration values. Therefore, the choice of MF_U as acceptability threshold aligns with established performance-criteria behaviour (Boylan and Russell, 2006; Chemel et al., 2010) with the advantage of not introducing any additional free parameters, and it can be applied to all pollutants for which measurement uncertainty parameters are set.

1.3. Threshold exceedances indicators

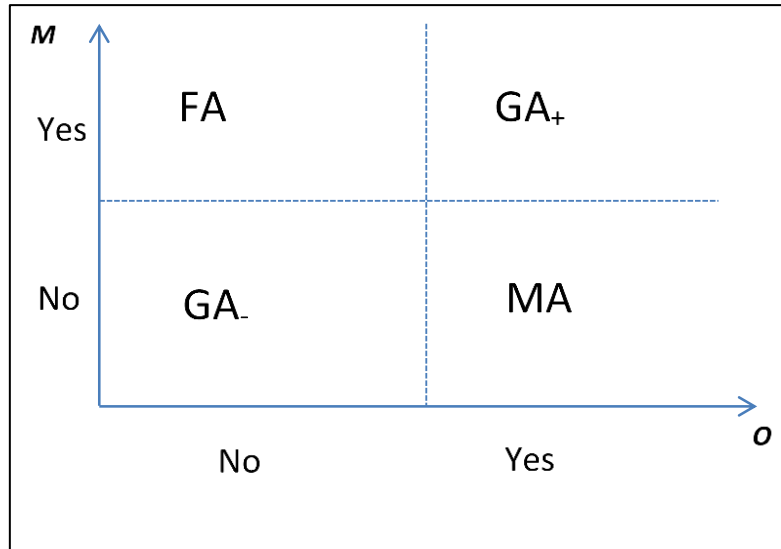
Forecasting system for short-term actions are required to be skilled in predicting the exceedance of a specific threshold e.g. the daily limit values set by European AAQDs. For decision makers the categorical answer yes/no is often the trigger that initiates the action so that it is of utmost importance to take it into account. The most used threshold indicators are included in the proposed validation approach, based on the 2x2 contingency table representing the joint distribution of categorical events (below/above the threshold value) predicted by the model (M) and observed by the measurements (O) as presented in Figure 1.

In this framework, four quantities can be defined:

- False Alarms (FA): Model values are above the threshold but not the observations
 - $M > \text{threshold} \ \& \ O \leq \text{threshold}$
- Missed Alarms (MA): Model values are below the threshold but observed values are above it
 - $M \leq \text{threshold} \ \& \ O > \text{threshold}$
- Good values below threshold (GA-): both model and observation are below or equal to the threshold.
 - $M \ \& \ O \leq \text{threshold}$
- Good values above threshold (GA+): both model and observations are above the threshold.
 - $M \ \& \ O > \text{threshold}$

Metrics derived from these quantities do not account for measurement uncertainty.

Figure 1: Schematic outline of the threshold exceedance quantities GA_+ , GA_- , FA and MA . The threshold is indicated by the dashed line.



Source: JRC

The counted alarms $CA = 'GA_+' + 'MA'$ include all cases where $O > \text{threshold}$. For a good forecast both FA and MA are small compared to GA_+ and GA_- .

Based on these quantities, the most used threshold indicators are included in the proposed validation approach. They are listed in Table 2, along with their formulation. All metrics values range from 0 to 1, with 1 being the optimal value.

Table 2. Categorical metrics included in the validation protocol.

Indicator	Formulation
Accuracy	$ACC = \frac{GA_+ + GA_-}{FA + GA_+ + GA_- + MA}$
Success Ratio	$SR = \frac{GA_+}{FA + GA_+}$
Probability of Detection	$POD \text{ (or } PD) = \frac{GA_+}{GA_+ + MA}$
FBias score	$FBIAS \text{ (or } FB) = \frac{FA + GA_+}{GA_+ + MA}$
Threat score	$TS = \frac{GA_+}{FA + GA_+ + MA}$
Gilbert Skill score	$GSS = \frac{GA_+ - H}{FA + GA_+ + MA - H}$

	$\text{with } H = \frac{(GA_+ + MA)(FA + GA_+)}{FA + GA_+ + GA_- + MA}$
--	---

Source: JRC

1.4. Assessment of capability to predict air quality indices

One of the main objectives of a forecasting system is to provide citizens with simple information about local air quality and its potential impact on their health, with special regard for sensitive and vulnerable groups. Air quality indices (AQIs) are designed to provide information on the potential effects of the different pollutants on people's health by means of a classification of concentration values in terms of qualitative categories. The AQI outcome is commonly provided by operational forecasting systems; therefore, its assessment has been included in the proposed validation approach, by means of a simple multiple threshold assessment, and the number of days predicted by the forecast model in each category is compared with the corresponding number of measured days.

The performance assessment depends on the chosen classification table/scheme. Several AQI tables are available, e.g. EEA (<https://airindex.eea.europa.eu/AQI/index.html>), UK ([https://uk-air.defra.gov.uk/air-pollution/daqi](https://uk-air.defra.gov.uk/air-pollution/daqi;); <https://uk-air.defra.gov.uk/air-pollution/daqi?view=more-info>) and US EPA (<https://www.airnow.gov/aqi/aqi-basics/>).

2. Forecasting indicators: graphical implementation proposal

A number of specific forecast diagrams are proposed in the following that reflect the evaluation framework described above.

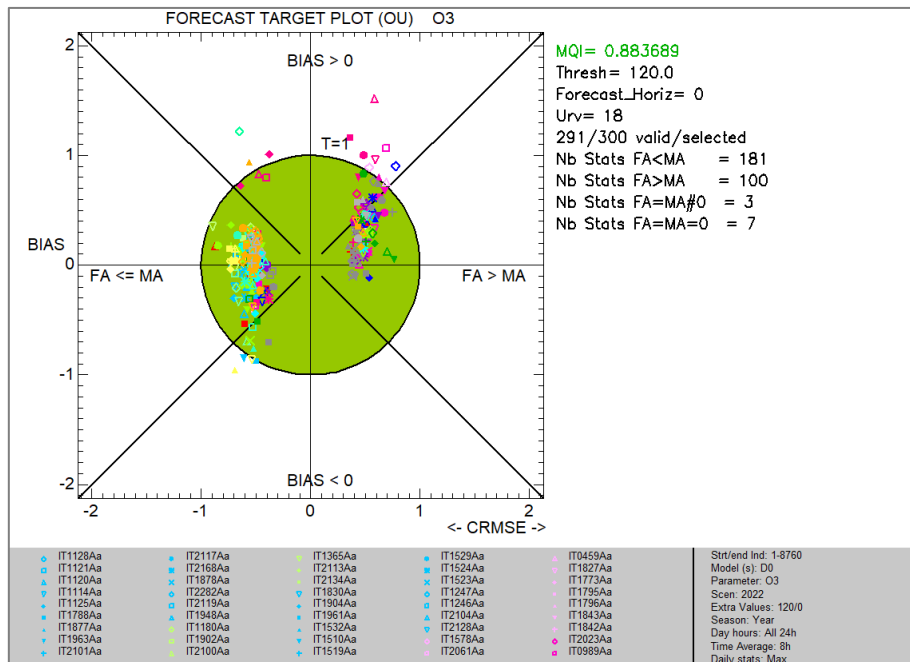
2.1. Forecast Target Plot

An example of a Forecast Target Plot is provided in Figure 2. In this plot, developed based on the well-established Assessment Target Plot (Jannsen and Thunis, 2022), MQI_f (i.e., $RMSE/RMSE_{persistence}$) represents the distance between the origin and a given point. For each point (representing one station), the ordinate is $BIAS/RMSE_{persistence}$, the abscissa is $CRMSE/RMSE_{persistence}$. The green area identifies the fulfilment of MQO_f at each monitoring station. The upper and lower quadrants identify positive and negative BIAS, respectively. Because CRMSE is always positive by definition, only the right part of the diagram is strictly needed in the Forecast Target plot. Nevertheless, the negative x-axis section is used to provide additional information, i.e. to compare False Alarm (FA) vs Missed Alarm (MA). In more detail: FA/MA ratio is considered to differentiate the right and left parts of the target diagram, as follows

$$\text{If } \frac{FA}{MA} \leq 1 \Rightarrow \text{Left}$$

$$\text{If } \frac{FA}{MA} > 1 \Rightarrow \text{Right}$$

Figure 2: Forecast Target plot. The stations are represented by various symbols. The MQI_f value associated with the 90th percentile worst station is reported on the right, together with parameters used for the computation and some statistics concerning categorical metrics.



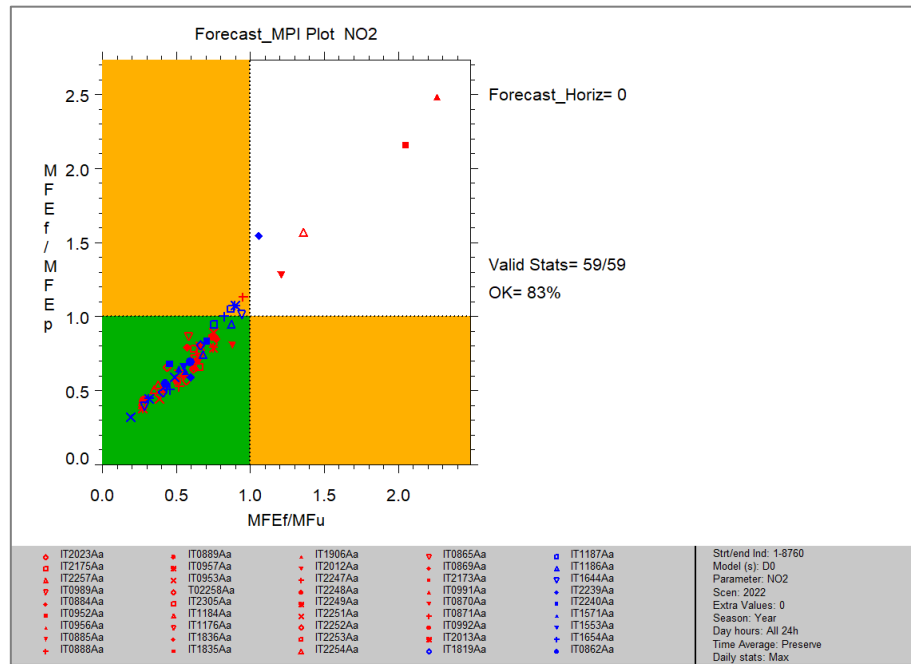
Source: JRC

2.2. Forecast MPI Plot

In addition to the MQI_f , the MPIs defined in Section 1.2 help interpreting the comparison of forecast model performances with persistence model ones. The fulfilment of the MPC defined in Table 1 is provided by the Forecast MPI Plot (Figure 3).

In this plot, forecast performance (MFE_f) is compared to Mean Fractional Uncertainty (MF_U) along the x-axis (Equation 7) and to the persistence model performance (MFE_p) on the y-axis (Equation 6). The green area identifies the area of fulfilment of both criteria, in the orange areas fall the stations fulfilling only one.

Figure 3: Forecast MPI plot. Forecast horizon value is reported together with the fraction of valid stations, and the percentage of stations where both the MPCs are fulfilled.



Source: JRC

2.3. Forecast Threshold Performance Plot

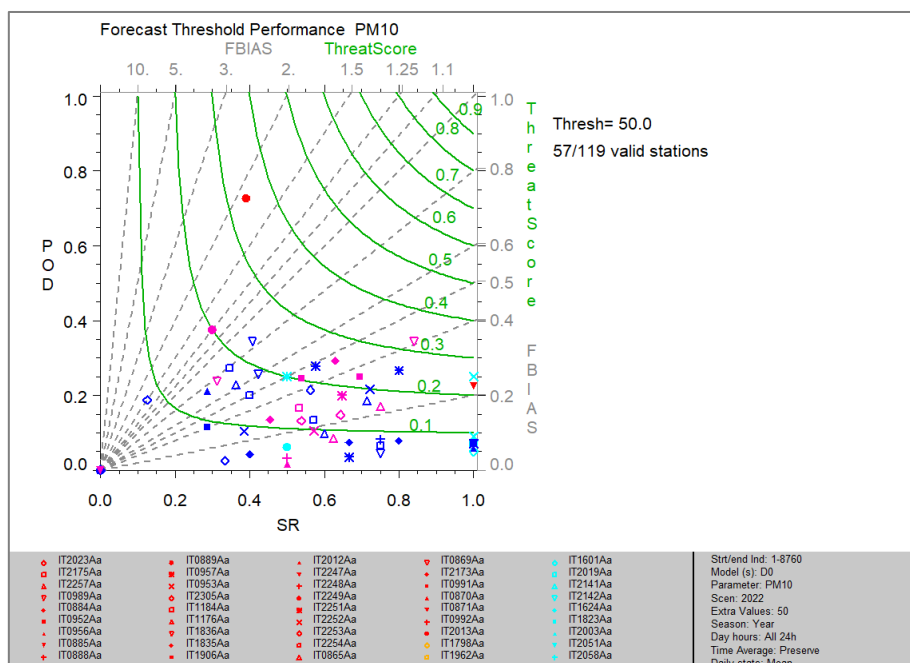
In addition to the comparison with the persistence model, indicators related to threshold exceedances are also required to evaluate the performance of the forecast model (Section 1.3). The four forecast indicators POD (Probability of Detection), SR (Success Ratio), FBias (FBias score) and TS (Threat score) can be put into one Forecast Threshold Performance Plot (Figure 4).

This plot uses SR values on the x-axis and POD values on the y-axis. Since FBias and TS indicators are related to POD and SR, they are represented by additional isolines. Good forecasts with high POD and SR are situated in the upper right corner.

The Forecast Threshold Performance Plot alone does not indicate whether a forecast model is adequate. Therefore, POD and SR values obtained with the persistence model are used also in this circumstance as a benchmark. This normalisation gives rise to the Forecast Threshold Normalized Performance Plot (Figure 5). The green area represents forecasts with better POD and SR threshold indicators than the persistence model. In the white zone, the model performs worse than the

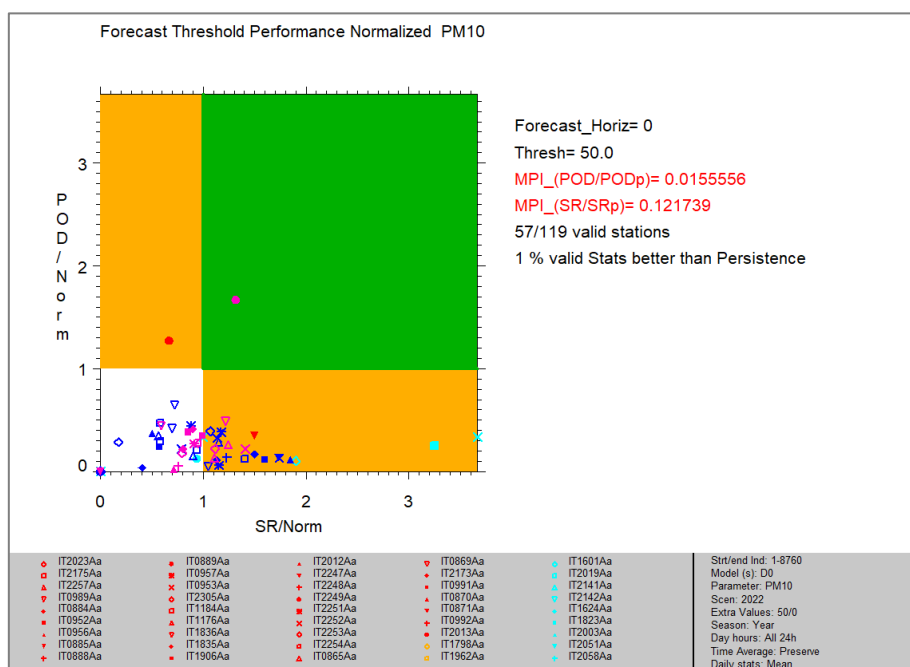
persistence model on both indicators. In the orange zone, one of the two indicators is better than the benchmark.

Figure 4: Forecast Threshold Performance Plot including the indicators POD, SR, FBIAS, and TS.



Source: JRC

Figure 5: Forecast Threshold Normalized Performance Plot as a normalised version of the Forecast Performance Diagram. POD and SR values obtained with the persistence model are used as benchmark and normalisation factor for the X and Y axis. The 10th percentile values of the normalised POD and SR values (i.e. POD/POD_p and SR/SR_p) are reported on the right as indicative quality indicators.



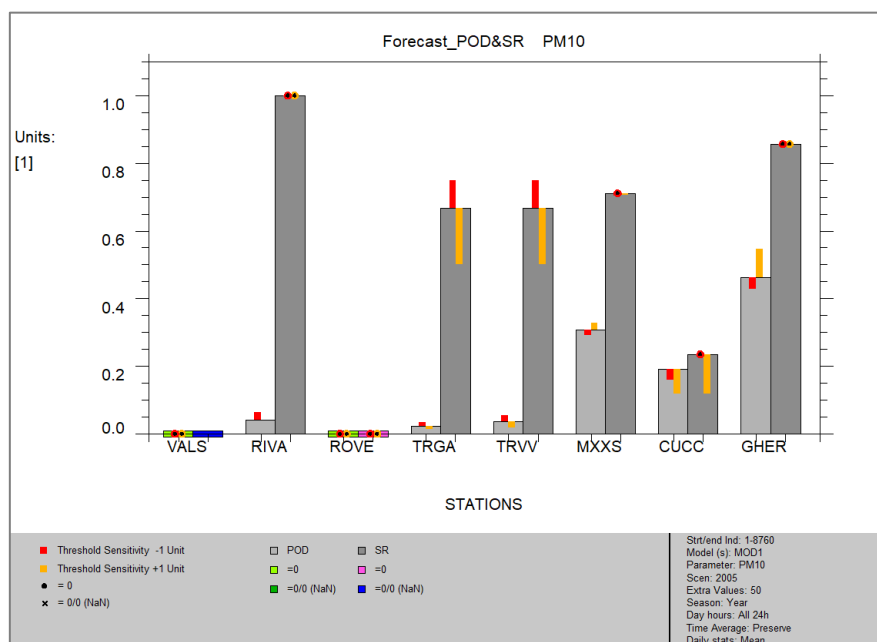
Source: JRC

2.4. Diagrams for exceedance indicators

The prediction of fixed threshold exceedances is a critical aspect for a deterministic model, whereas for probabilistic forecasting models, such as ensemble models, Bayesian statistical models or artificial intelligence models, it is possible to quantify the probability of a threshold exceedance thus providing a more comprehensive indication to decision makers. Furthermore, in many European areas, the exceedance of the regulatory threshold is a rare event compared to the non-exceedance case. In categorical statistical scores, the event of interest is the less probable one, and consequently, an incorrect classification of few cases penalizes the performance. For this reason, an evaluation of sensitivity of model performance to variation of threshold values is recommended.

Figure 6 presents a bar plot for POD (light grey) and SR (dark grey) for each station, showing how these indicators vary with changes in the threshold value. The red bar indicates the change in the indicator when the threshold is reduced by 1 unit, the yellow bar the change when the threshold is increased by 1 unit.

Figure 6: POD & SR diagram. POD (light grey) and SR (dark grey) values are given for each station. The sensitivity with respect to the threshold value is expressed by the red and yellow bars. Values of 0 (zero) and O/O (NaN) are indicated by different colours.



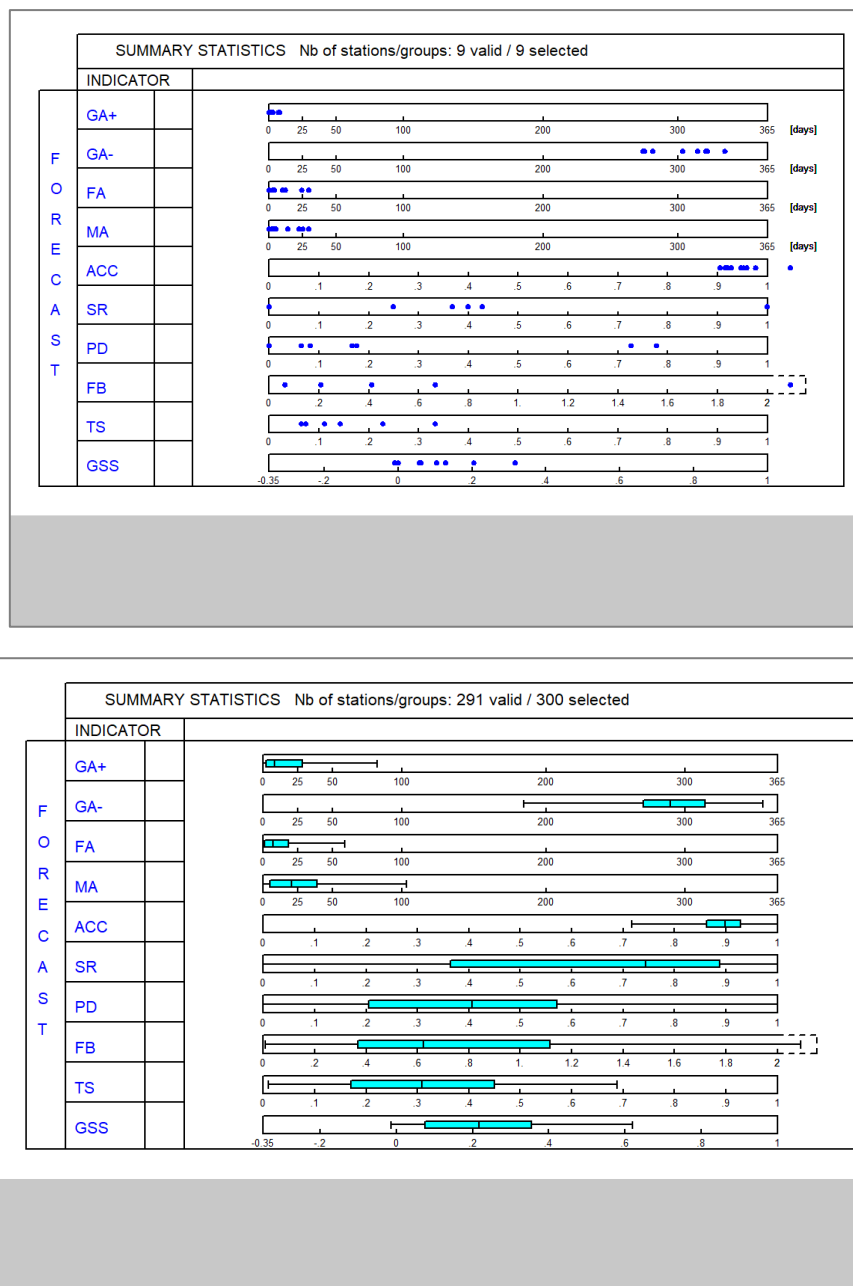
Source: JRC

2.5. Forecast Summary Report

The outcomes of all the exceedance indicators are summarized in the Forecast Summary Report.

An example of a Forecast Summary Report is presented in Figure 7. A different graphical layout is applied depending on the number of stations taken into account in the analysis: if the number of stations is below 15, each of the dots represents a station for which the forecast indicators are evaluated (top); if the number of stations is above or equal 15, boxplots are used to represent the statistical distribution of the indicators values (bottom).

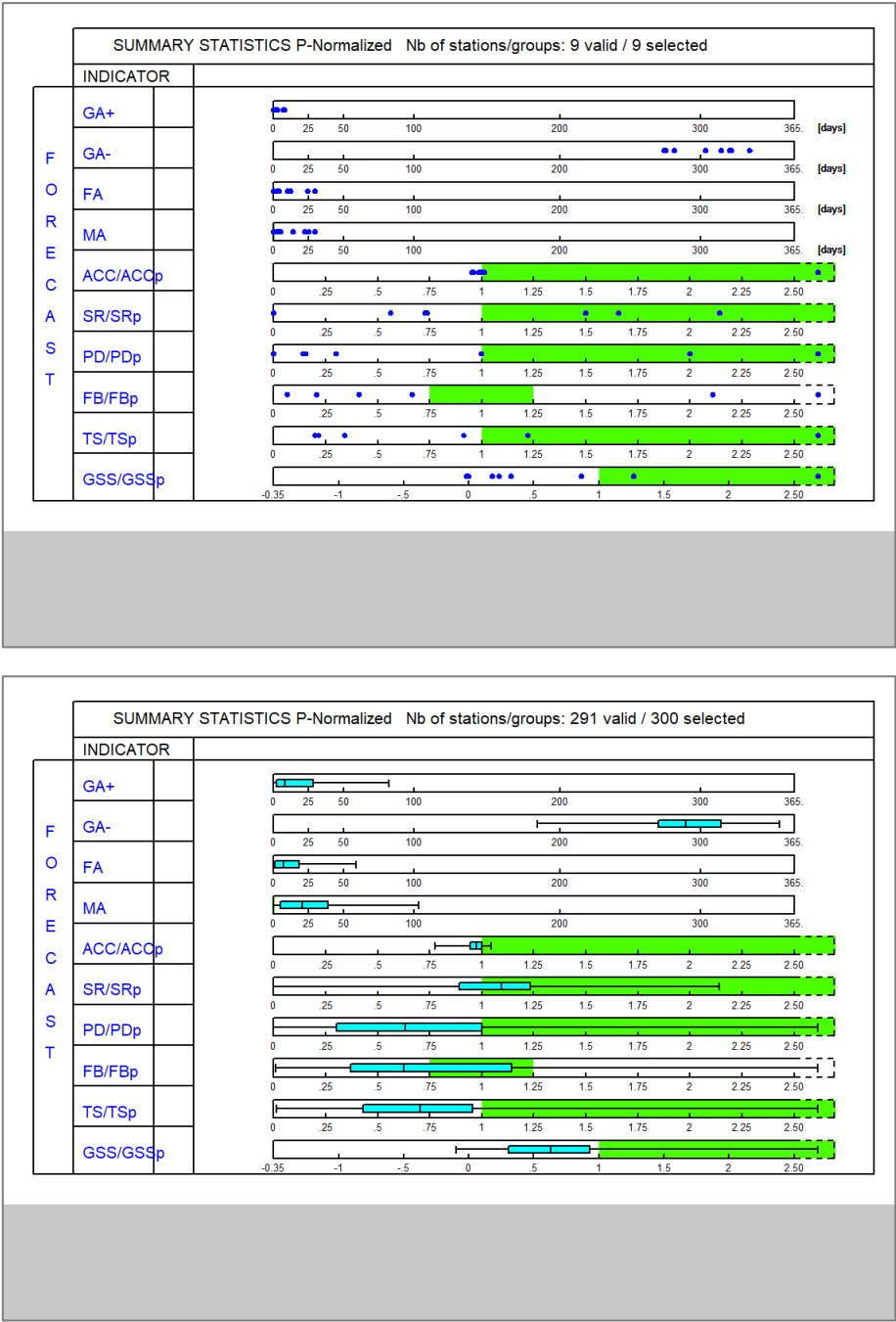
Figure 7: Forecast Summary Report, when the number of stations is below 15 (top) and when the number of stations is above or equal 15 (bottom).



Source: JRC

To determine whether a forecast model is adequate, indicators values obtained with the persistence model can serve as a benchmark. A normalized version of the Forecast Summary Report can be produced, where the “adequate” zone is shaded green, indicating that the model performs better than the persistence model for that particular indicator. An example of the Forecast Summary P-Normalized Report is provided in Figure 8 when the number of stations is below 15 (top) and when the number of stations is above or equal 15 (bottom).

Figure 8: Forecast Summary P-Normalized Report, when the number of stations is below 15 (top) and when the number of stations is above or equal 15 (bottom).



Source: JRC

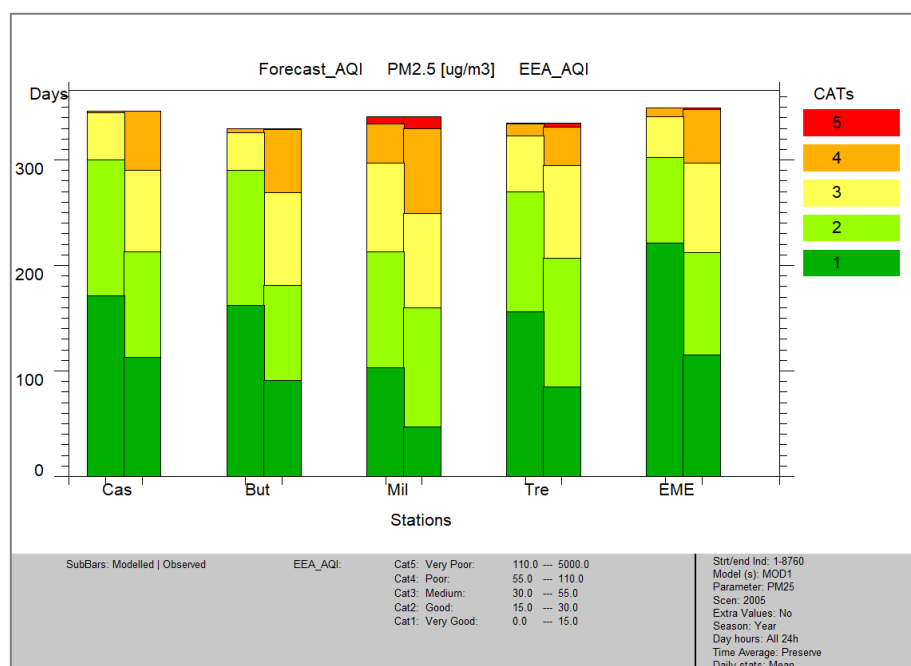
2.6. Forecast diagrams using Air Quality Indices

When model performance in predicting AQI values is assessed by comparing the number of days in each AQI category between observed and predicted data, results can be visualised using a bar plot (Figure 9).

In this plot, two paired bars are shown for each station: one representing forecast values and one representing observed values. Each bar indicates the number of days that fall within a given

air-quality category. The corresponding AQI classification table appears in the shaded grey area below, with associated colour codes displayed on the right-hand side of the figure.

Figure 9: Forecast Air Quality Index diagram showing the number of days in each index category. Two bars (Model and Observation) per station.



Source: JRC

2.7. Overview of the metrics

Table 3 lists the metrics included in the validation protocol together with their key features, specifically:

- if they assess overall model performance or the capability of correctly providing the exceedances of concentration thresholds;
- if they account for measurement uncertainty;
- whether a quality criterion is proposed, and if so, which one.

Table 3. Overview of the metrics included in the validation protocol.

Metric	Type of evaluation	Is measurement uncertainty accounted for?	Quality criterion
MQI_f	overall model performance	yes	$MQI_f \leq 1$
MPI_1	overall model performance	yes	$MPI_1 \leq 1$
MPI_2	overall model performance	yes	$MPI_2 \leq 1$

<i>ACC</i>	capability of correctly providing the exceedances	no	none
<i>SR</i>	capability of correctly providing the exceedances	no	none
<i>POD</i>	capability of correctly providing the exceedances	no	none
<i>FBIAS</i>	capability of correctly providing the exceedances	no	none
<i>TS</i>	capability of correctly providing the exceedances	no	none
<i>GSS</i>	capability of correctly providing the exceedances	no	none

3. Conclusions and open issues

This guidance document presents a standardized procedure for evaluating air quality forecast applications. Following FAIRMODE discussions on specific challenges in forecasting, a common benchmarking framework has been developed to support model developers and users under the European Ambient Air Quality Directives (AAQDs).

The proposed approach uses a reference forecast (the persistence model) as a benchmark, explicitly includes measurement uncertainty for some metrics, and defines objective performance criteria to determine whether a forecast application reaches an acceptable quality level for regulatory use. Additional indicators complement these criteria to better analyse model strengths and weaknesses.

By relying on a harmonised validation protocol, this framework enables the comparison of performances across different forecasting systems, supporting transparency and consistency in the implementation of the European AAQDs.

Applications of the methodology within the FAIRMODE community—covering various models, pollutants, and spatial scales across Europe—have confirmed its general usefulness but also highlighted areas for improvement.

First, the current use of the **Forecast Modelling Quality Objective (MQO_r)** as a pass–fail test may need to be reconsidered in light of Directive (EU) 2024/2881, which emphasises forecasting as a tool for developing short-term action plans. Fitness for purpose should therefore focus more on the model’s ability to represent high-concentration episodes and correctly predict exceedances, rather than on overall average performance. In this context, MQI_r could remain a complementary metric, rather than the main one.

Second, community tests have shown that even advanced forecasting systems often struggle to outperform the persistence model in **predicting exceedances**. Alternative or refined approaches should therefore be explored to define “fitness for purpose” in this domain without relying solely on persistence-based benchmarking. In this context, the feasibility of incorporating ensemble, probabilistic forecasting models or other approaches remains an open question.

Third, the assessment of the **Air Quality Index (AQI)** forecasting capability remains the most open issue. Accurate and timely AQI predictions are becoming increasingly important for public information duties under the new legislation. The current cumulative approach does not assess the correct timing of AQI levels, and no criteria are yet defined to determine acceptable AQI forecast quality. Further benchmarking exercises are needed to establish reference performance levels.

Fourth, the overview and experience of Member States in establishing and implementing short-term action plans is a new open issue that haven’t been discussed yet. In the AAQD, those plans are mentioned and associated with exceedances, that could be forecasted by the models – especially in terms of extent and duration. Review of what kind of short-term action could be taken upon forecasted exceedance, its extent and effectiveness should be done in upcoming years.

These open issues will continue to be discussed within the FAIRMODE community to stimulate additional research, promote inter-comparison exercises, and guide the progressive improvement of the current validation protocol.

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List of abbreviations and definitions

AQ	Air Quality
AQI	Air Quality Index
ACC	ACCuracy score (forecast)
AAQD	Ambient Air Quality Directive
CA	Counted Alarms (forecast)
CRMSE	Centered Root Mean Square Error
EEA	European Environmental Agency
EU	European Union
FA	False Alarms (forecast)
FAIRMODE	EU Forum for AIR quality MODElling
FBIAS	Fractional bias
FH	Forecast Horizon
GA	Good Alarms (forecast)
GSS	Gilbert Skill Score (forecast)
M	Model
MA	Missed Alarms (forecast)
MFE	Mean Fractional Error
MF _u	Mean Fractional Uncertainty
MPC	Modelling Performance Criteria
MPI	Modelling Performance Indicator
MQI _f	Forecast Modelling Quality Indicator
MQO _f	Forecast Modelling Quality Objective
NO ₂	Nitrogen Dioxides
O	Observations
O ₃	Ozone
PM	Particulate Matter
POD	Probability Of Detection
RMSE	Root Mean Square Error
SR	Success Ratio (forecast)
TS	Threat Score (forecast)

U	Measurement Uncertainty
UK	United Kingdom
USEPA	United States Environmental Protection Agency

List of figures

Figure 1: Schematic outline of the threshold exceedance quantities GA+, GA-, FA and MA. The threshold is indicated by the dashed line.	8
Figure 2: Forecast Target plot. The stations are represented by various symbols. The MQI_f value associated with the 90 th percentile worst station is reported on the right, together with parameters used for the computation and some statistics concerning categorical metrics.	10
Figure 3: Forecast MPI plot. Forecast horizon value is reported together with the fraction of valid stations, and the percentage of stations where both the MPCs are fulfilled.	11
Figure 4: Forecast Threshold Performance Plot including the indicators POD, SR, FBIAS, and TS.	12
Figure 5: Forecast Threshold Normalized Performance Plot as a normalised version of the Forecast Performance Diagram. POD and SR values obtained with the persistence model are used as benchmark and normalisation factor for the X and Y axis. The 10 th percentile values of the normalised POD and SR values (i.e. POD/POD_p and SR/SR_p) are reported on the right as indicative quality indicators.	12
Figure 6: POD & SR diagram. POD (light grey) and SR (dark grey) values are given for each station. The sensitivity with respect to the threshold value is expressed by the red and yellow bars. Values of 0 (zero) and 0/0 (NaN) are indicated by different colours.	13
Figure 7: Forecast Summary Report, when the number of stations is below 15 (top) and when the number of stations is above or equal 15 (bottom).	14
Figure 8: Forecast Summary P-Normalized Report, when the number of stations is below 15 (top) and when the number of stations is above or equal 15 (bottom).	15
Figure 9: Forecast Air Quality Index diagram showing the number of days in each index category. Two bars (Model and Observation) per station.	16

List of tables

Table 1. Model performance indicators and criteria for MFE.....	7
Table 2. Categorical metrics included in the validation protocol.....	8
Table 3. Overview of the metrics included in the validation protocol.....	16

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